

Paddy Crop Analysis Using Python

1. Project Overview

The **Paddy Crop Analysis Using Python** project aims to analyse a paddy cultivation dataset using Python to identify factors that influence crop yield. The project includes data cleaning, preprocessing, exploratory data analysis (EDA), statistical analysis, data visualization, and predictive suggestions. Using libraries such as **Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn**, the project transforms raw agricultural data into meaningful insights to support better farming decisions and improve paddy productivity.

2. Data Source

- **Dataset Name:** Paddy Dataset
- **Source:** UCI Machine Learning Repository – Paddy Dataset (<https://archive.ics.uci.edu/dataset/1186/paddy+dataset>)
- **File Type:** CSV (.csv)
- **Number of Records:** 2,789
- **Number of Attributes:** 45

The dataset contains information related to paddy cultivation, including crop variety, soil type, seed rate, fertilizer usage, weather conditions, irrigation, pest management, and yield. It is used to analyse the factors affecting paddy production and generate meaningful insights using Python.

3. Problem Statement

Paddy yield is influenced by several factors, including soil type, seed rate, fertilizer application, weather conditions, irrigation, and pest management. Analysing these factors manually is difficult and time-consuming. This project aims to analyse the paddy dataset using Python to identify patterns, understand the factors affecting crop yield, and generate insights that support better agricultural planning and decision-making.

4. Objectives

- To understand the structure of the paddy dataset.
 - To clean and preprocess the data for analysis.
 - To perform Exploratory Data Analysis (EDA).
 - To identify the factors affecting paddy yield.
 - To visualize patterns and relationships using Python.
 - To build a predictive suggestion for paddy yield estimation.
 - To generate meaningful insights for better agricultural decision-making.
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5. Attribute (Column / Feature) Details

Attribute Name	Description
Hectares	Total cultivated land area under paddy cultivation
Agriblock	Agricultural block/region where the field is located
Variety	Variety of paddy cultivated
Soil Types	Type of soil used for cultivation
Seedrate (Kg)	Quantity of seeds used for cultivation (in kilograms)
Nursery	Nursery preparation details
Nursery Area (Cents)	Area allocated for nursery cultivation
DAP_20days	Quantity of DAP fertilizer applied after 20 days
Urea	Quantity of urea fertilizer applied
Potash	Quantity of potash fertilizer applied
Irrigation	Irrigation method or frequency used

6. Tools & Technologies

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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7. Data Pre-Processing and Feature Engineering

The dataset was prepared for analysis using data preprocessing techniques.

Data Cleaning Steps

- Checked data types of all columns
- Identified missing values
- Handled null values
- Removed duplicate records

- Corrected inconsistent data
- Converted data types where required

Feature Engineering

New features were created to improve analysis.

Examples:

- Age groups were created from the Age column
 - Purchase categories were created based on Total Spend
 - Additional calculated fields were generated where required
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8. Analysis and Visualizations

The cleaned dataset was analysed using **Python** to understand the characteristics of paddy cultivation data. **Exploratory Data Analysis (EDA)** and statistical analysis were performed to identify patterns, trends, and relationships among the variables. Various visualizations, including histograms, box plots, count plots, scatter plots, heatmaps, pair plots, and violin plots, were created using **Matplotlib** and **Seaborn** to explore the data and present meaningful insights into the factors affecting paddy cultivation and yield.

9. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure and characteristics of the paddy dataset.

Initial inspection included:

- Data frame structure verification
- Statistical summary of numerical columns
- Missing value verification
- Duplicate record validation
- Data type verification
- Distribution analysis of numerical and categorical variables

EDA helped in understanding the distribution, patterns, and characteristics of the data, providing a strong foundation for further analysis.

10. Statistical Analysis

Statistical analysis was performed to summarize and understand the numerical characteristics of the paddy dataset.

Statistical measures included:

- Mean
- Median
- Mode
- Minimum value
- Maximum value
- Standard deviation
- Variance

These measures helped to understand the central tendency, variability, and overall distribution of the data, providing valuable insights into the characteristics of the paddy cultivation variables.

11. Data Visualization

Data visualization techniques were used to represent the paddy dataset graphically and make patterns, trends, and relationships easier to understand. Various charts created using **Matplotlib** and **Seaborn** helped analyse the distribution of variables, compare categories, identify outliers, and examine relationships among agricultural factors affecting paddy cultivation.

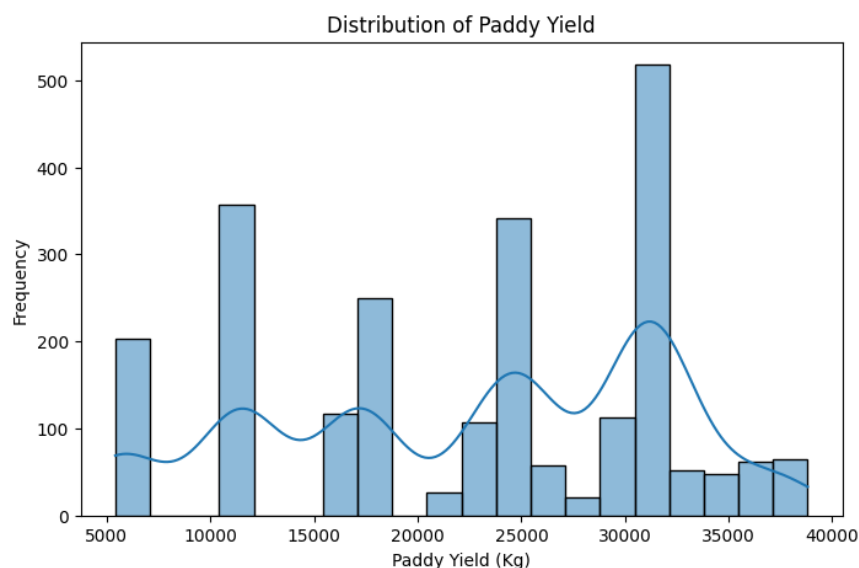
1. Univariate Analysis

Univariate analysis was performed to examine the distribution of individual variables in the paddy dataset.

Analysis Performed:

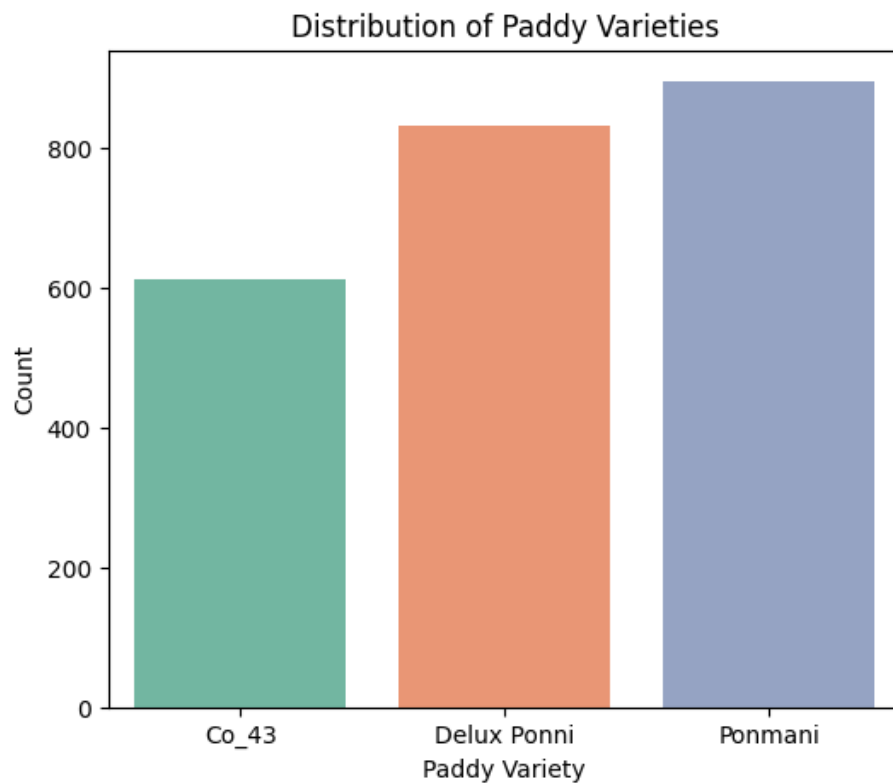
1. Histogram – Distribution of Paddy Yield (with image)

Interpretation: The histogram shows the distribution of paddy yield values, helping to understand the spread, frequency, and overall pattern of crop production



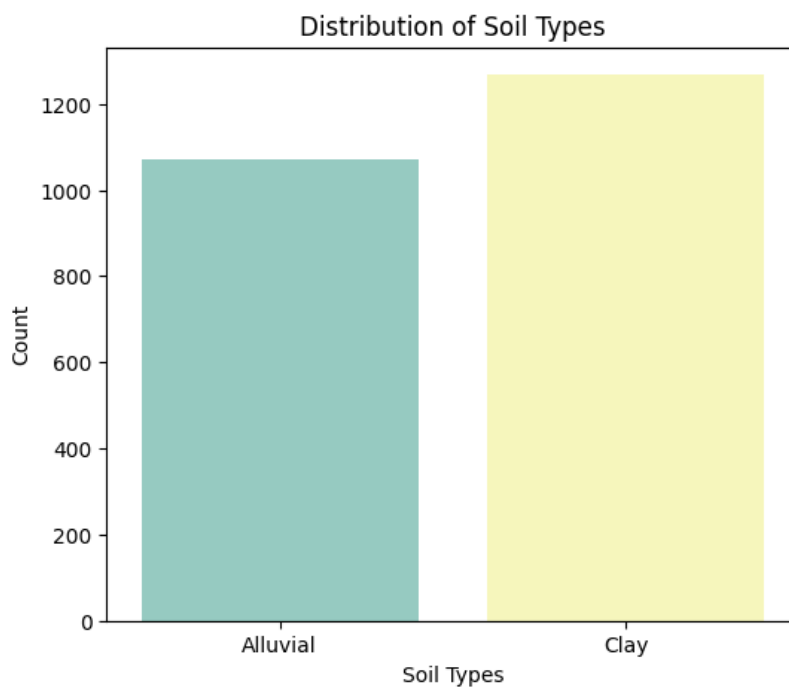
2. Count Plot – Paddy Variety (with image)

Interpretation: The count plot compares the frequency of different paddy varieties and identifies the most commonly cultivated variety in the dataset.



3. Count Plot – Soil Types (with image)

Interpretation: The count plot shows the distribution of soil types used for paddy cultivation, highlighting the most prevalent soil type in the dataset.



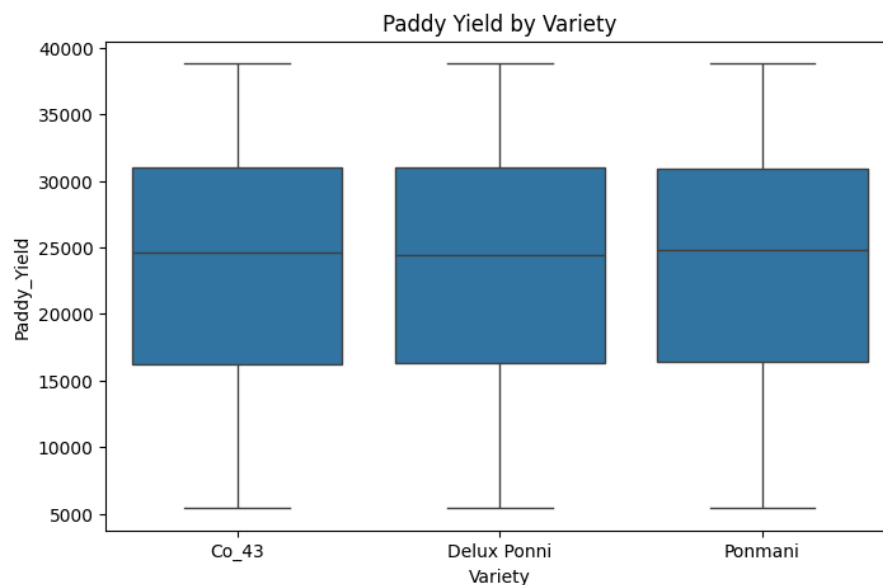
2. Bivariate Analysis

Bivariate analysis was performed to examine the relationship between two variables and understand how different factors influence paddy yield.

Analysis Performed:

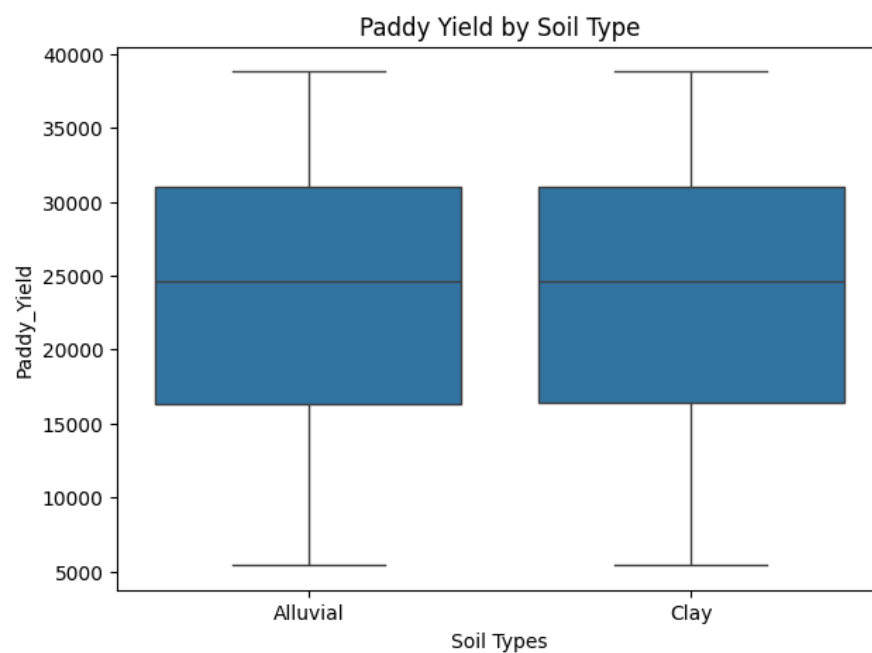
1. Box Plot – Variety vs Paddy Yield (with image)

Interpretation: The box plot compares the distribution of paddy yield across different varieties, highlighting variations in yield performance and the presence of outliers.



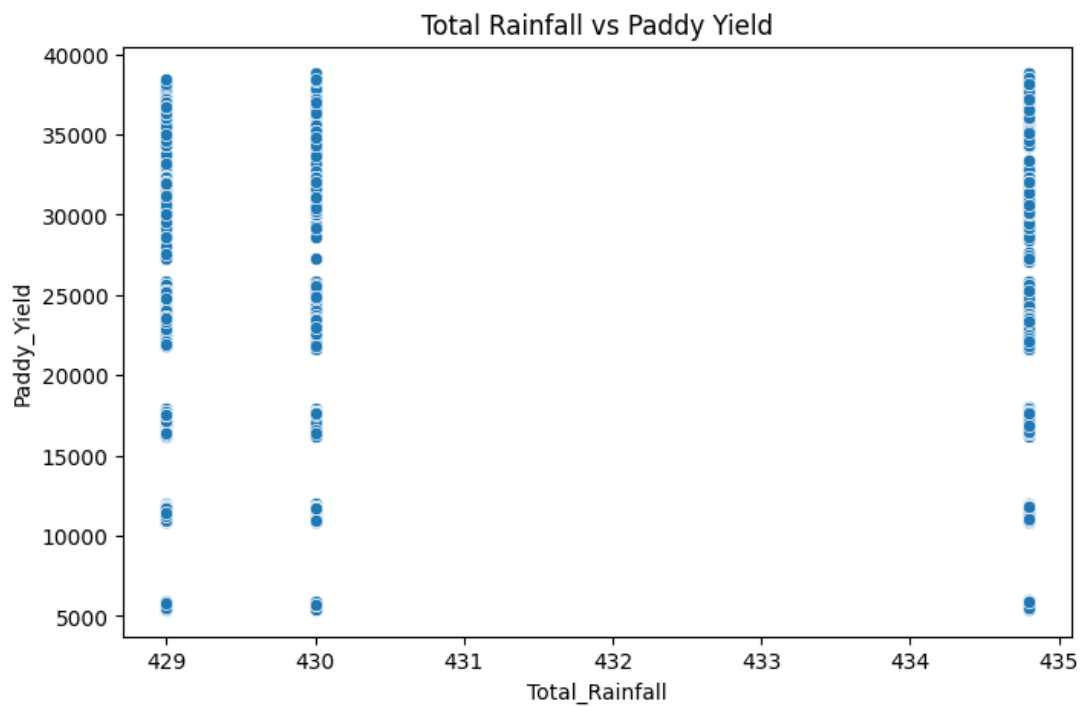
2. Box Plot – Soil Type vs Paddy Yield (with image)

Interpretation: The box plot shows how paddy yield varies across different soil types, helping identify which soil type is associated with higher crop yield.



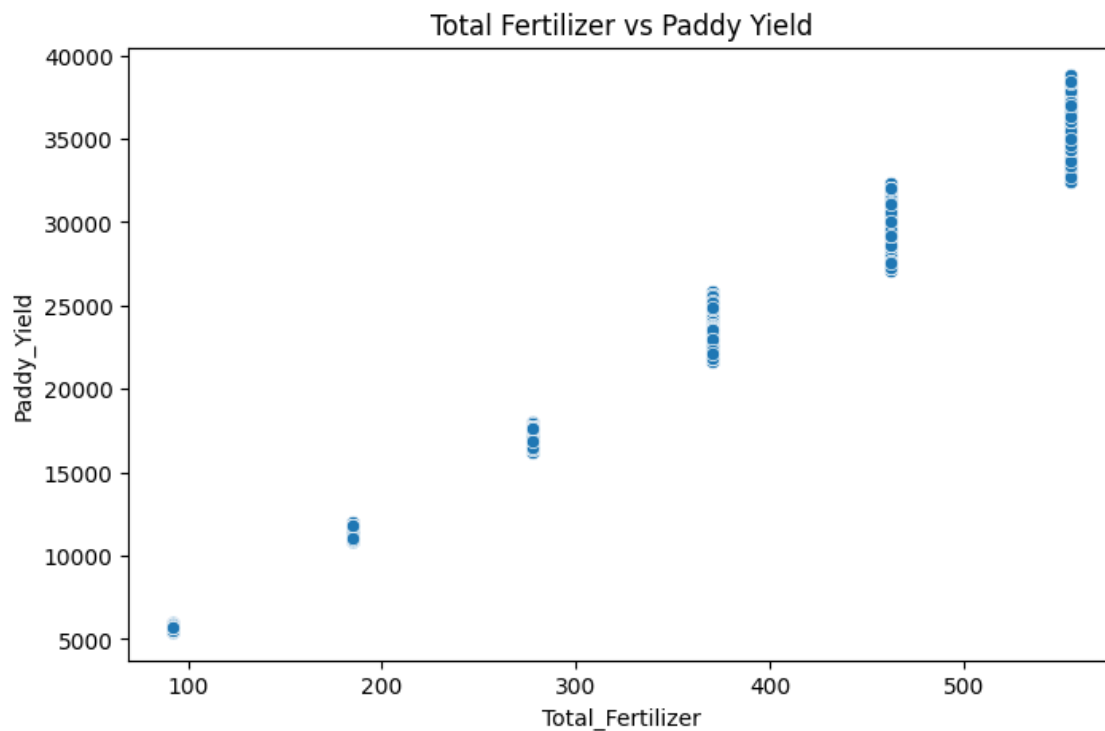
3. Scatter Plot – Total Rainfall vs Paddy Yield (with image)

Interpretation: The scatter plot illustrates the relationship between total rainfall and paddy yield, indicating whether rainfall has an influence on crop production.



4. Scatter Plot – Total Fertilizer vs Paddy Yield (with image)

Interpretation: The scatter plot examines the relationship between total fertilizer application and paddy yield, helping to understand how fertilizer usage affects crop production.



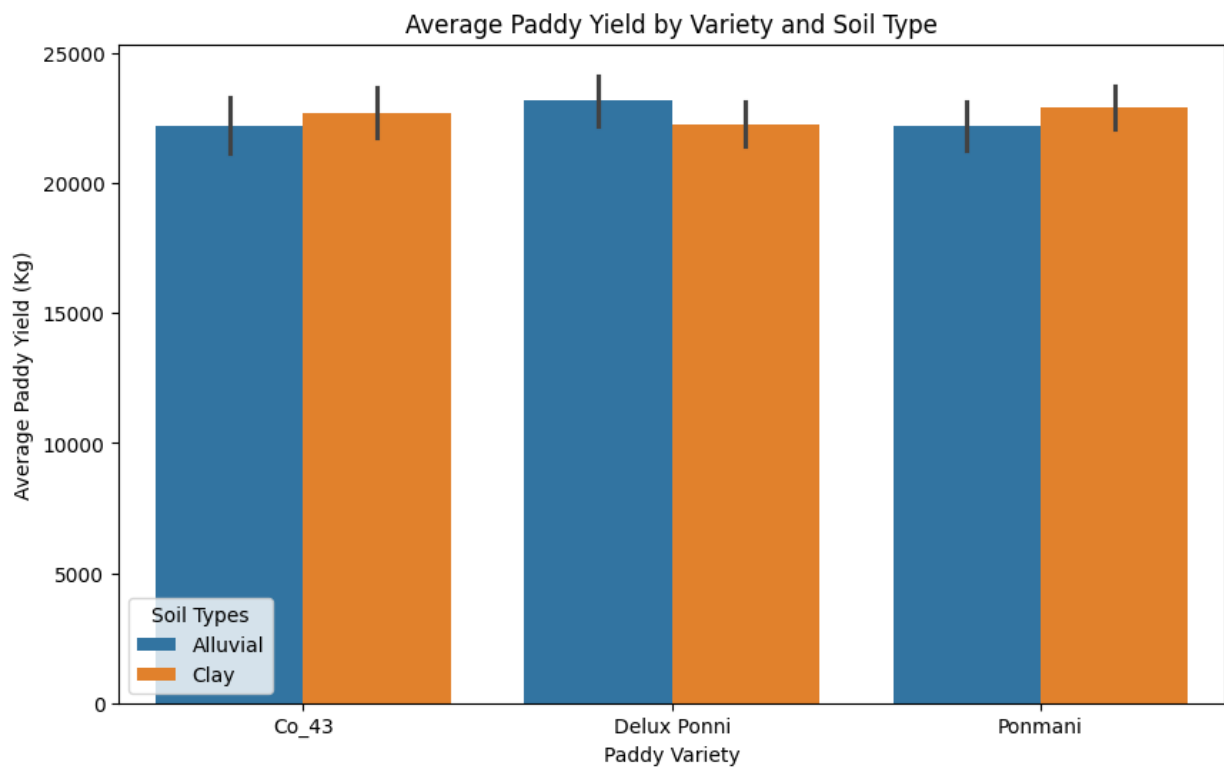
3. Multivariate Analysis

Multivariate analysis was performed to examine the relationship among multiple variables and understand their combined effect on paddy yield.

Analysis Performed:

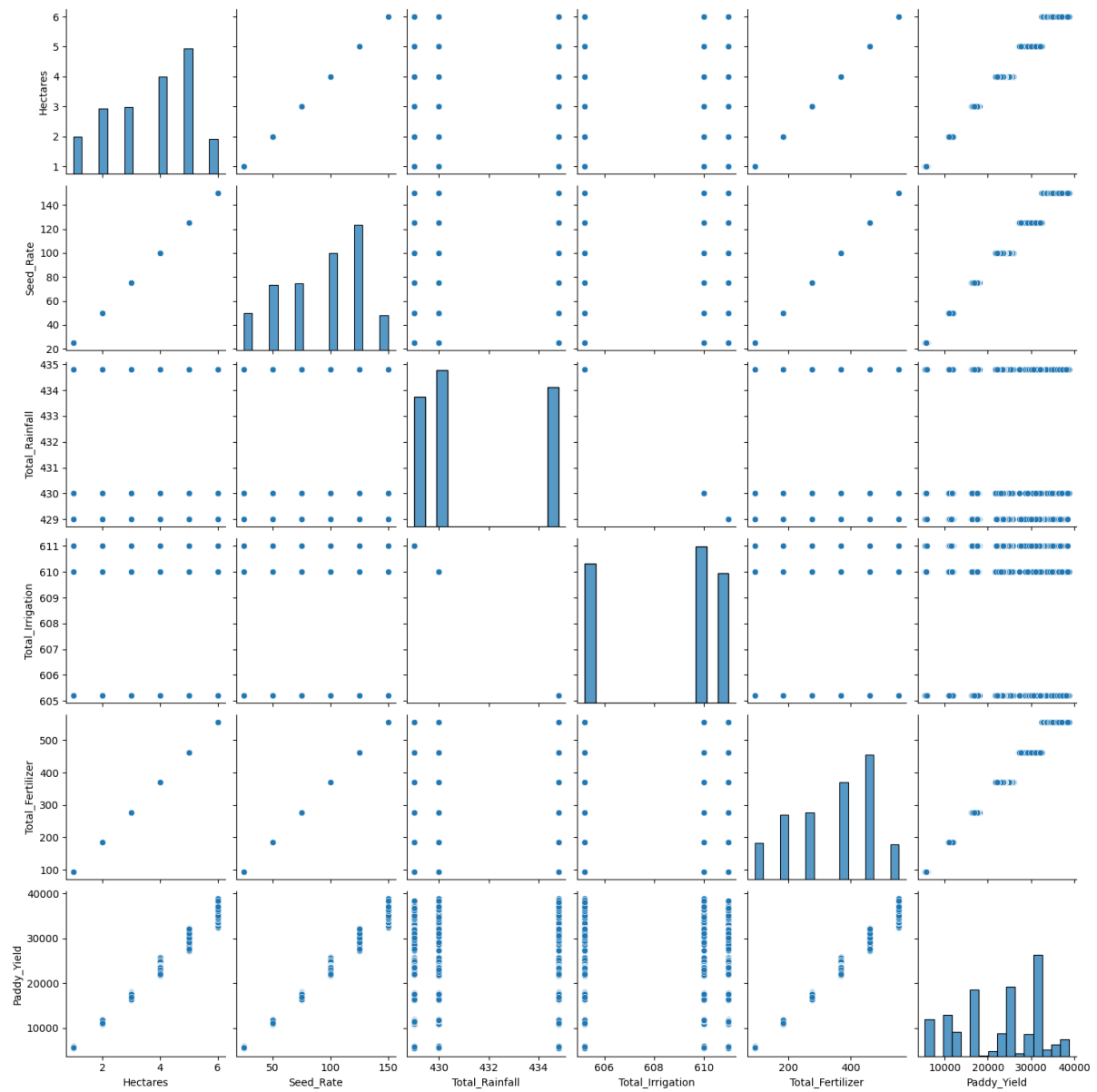
1. Grouped Bar Chart – Variety × Soil Type vs Paddy Yield (with image)

Interpretation: The grouped bar chart compares paddy yield across different combinations of paddy varieties and soil types, helping identify the best-performing combinations.



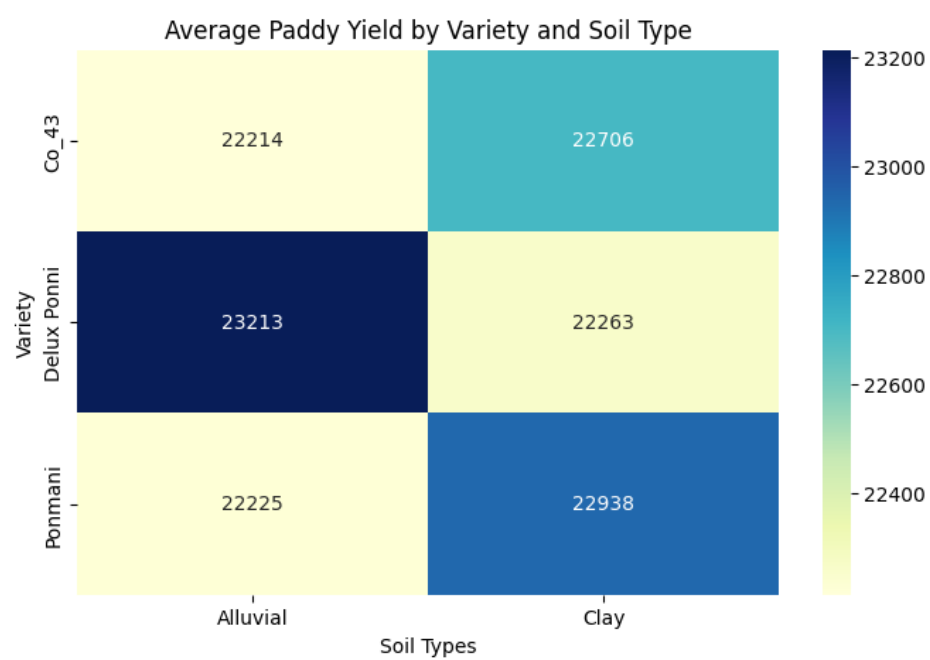
2. Pair Plot (with image)

Interpretation: The pair plot visualizes the relationships among multiple numerical variables, showing distributions, trends, and potential correlations between cultivation factors.



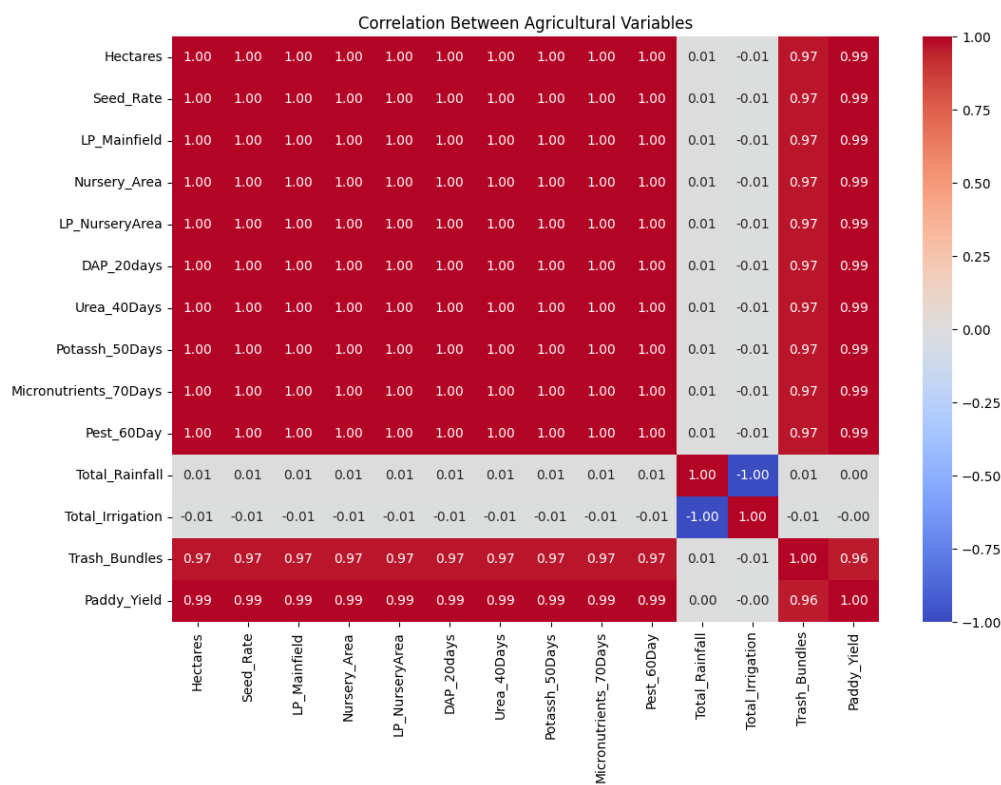
3. Pivot Table Heatmap (with image)

Interpretation: The pivot table heatmap displays the average paddy yield across combinations of key categorical variables, making it easier to identify high- and low-performing groups.



4. Correlation Heatmap (with image)

Interpretation: The correlation heatmap illustrates the strength and direction of relationships among numerical variables, helping identify factors that are positively or negatively associated with paddy yield.



12. Summary of Findings

- Paddy yield varies across different paddy varieties.
 - Soil type has a noticeable impact on paddy yield.
 - Rainfall shows an influence on crop production.
 - Fertilizer application is associated with variations in paddy yield.
 - Correlation analysis identified relationships among cultivation factors affecting paddy production.
 - Data visualization helped identify patterns, trends, and outliers in the dataset.
 - Exploratory Data Analysis (EDA) provided valuable insights into the characteristics of paddy cultivation data.
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13. Types of Analysis

1. Descriptive Analysis

Descriptive analysis summarizes and describes the characteristics of the paddy dataset.

Examples:

- Distribution of paddy yield
 - Count of paddy varieties
 - Distribution of soil types
 - Summary statistics of cultivation variables
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2. Diagnostic Analysis

Diagnostic analysis helps identify the factors influencing paddy yield.

Examples:

- Impact of paddy variety on yield
 - Effect of soil type on crop production
 - Relationship between rainfall and paddy yield
 - Effect of fertilizer application on paddy yield
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3. Predictive Analysis

Predictive analysis estimates possible outcomes based on historical data.

Examples:

- Identifying cultivation factors associated with higher paddy yield
 - Analysing relationships among agricultural variables to understand yield patterns
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4. Prescriptive Analysis

Prescriptive analysis provides recommendations based on the insights obtained from the data.

Examples:

- Selecting suitable paddy varieties for improved yield
 - Adopting appropriate soil and fertilizer management practices
 - Optimizing cultivation practices based on rainfall conditions
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14. Future Enhancements and Suggestions

- Include additional paddy cultivation data from different regions and seasons for more comprehensive analysis.
 - Perform advanced statistical analysis to gain deeper insights into crop production factors.
 - Develop interactive dashboards using Power BI or Tableau for better data visualization.
 - Automate the data cleaning, analysis, and reporting process using Python.
 - Incorporate real-time weather and agricultural data for more detailed analysis.
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15. Conclusion

The **Paddy Crop Analysis Using Python** project successfully analysed the paddy dataset using Python data analysis techniques. Data preprocessing, statistical analysis, and visualizations helped identify important patterns and relationships among cultivation factors affecting paddy yield. The insights obtained from this analysis can support better agricultural planning and informed decision-making to improve paddy cultivation practices.